



Haircutting robots: a mobile robotics perspective

Zhendai Huang , Ameer Tamoor Khan & Shuai Li

To cite this article: Zhendai Huang , Ameer Tamoor Khan & Shuai Li (15 Jun 2026): Haircutting robots: a mobile robotics perspective, International Journal of Systems Science, DOI: [10.1080/00207721.2026.2687894](https://doi.org/10.1080/00207721.2026.2687894)

To link to this article: <https://doi.org/10.1080/00207721.2026.2687894>



Published online: 15 Jun 2026.



Submit your article to this journal [↗](#)



Article views: 29



View related articles [↗](#)



View Crossmark data [↗](#)



Haircutting robots: a mobile robotics perspective

Zhendai Huang^a, Ameer Tamoor Khan^b and Shuai Li^{a,c}

^aFaculty of Information Technology and Electrical Engineering, University of Oulu, Oulu, Finland; ^bDepartment of Plant and Environment Science, University of Copenhagen, Copenhagen, Denmark; ^cVTT-Technical Research Centre of Finland, Oulu, Finland

ABSTRACT

The development of haircutting robots has gained increasing attention in personal grooming and healthcare applications. However, existing research remains fragmented, often addressing perception, hair modelling, and motion planning in isolation and lacking a unifying theoretical foundation. To bridge this gap, this review examines robotic haircutting from the perspective of mobile robotics, highlighting the shared principles of localisation, coverage planning, and control. Specifically, localisation establishes the spatial relationship between the cutting tool and the human head; coverage planning formulates trimming as a multi-pass path coverage problem under safety and aesthetic constraints; and control strategies support compliant, efficient, and personalised execution. Furthermore, learning-based approaches and techniques from autonomous systems are discussed as important enablers of adaptability and robustness. By mapping the conceptual and methodological connections between mobile robotics and haircutting robots, this paper provides a structured synthesis of current research and proposes a unified framework to guide future studies in this emerging field.

ARTICLE HISTORY

Received 19 October 2025
Accepted 31 May 2026

KEYWORDS

Robotic haircutting; mobile robotics; localisation; coverage path planning; autonomous systems

1. Introduction

Population aging, labour shortages in service industries, and the growing interest in contactless interaction have increased the demand for automation in personal grooming, especially after COVID-19 (McKinsey & Company, 2020). Within this context, robotic haircutting has attracted attention as a potential application of service robotics. It may improve service accessibility, hygiene, and consistency, while reducing the physical burden on human hair stylists. From a technical perspective, robotic manipulators have advanced in terms of motion accuracy, control reliability, and system integration, and have been widely applied in manufacturing automation, medical assistance, and precision manipulation tasks (Cepolina & Razzoli, 2024; Kyrarini et al., 2021; Moshayedi et al., 2026). In parallel, beyond traditional industrial manipulators, the rapid development of commercial collaborative robotic platforms and desktop robot arms has provided more accessible and flexible solutions for prototyping and experimental validation of low-cost robotic systems (Chen et al., 2025; Liu et al., 2024; Wang et al., 2025). Unlike many conventional

service tasks, however, robotic haircutting is inherently contact-rich, safety-critical, and appearance-sensitive. The robot must operate close to the human head, interact with deformable hair of diverse physical properties, and produce results that are judged not only by geometric completeness but also by aesthetic quality and personalisation. These characteristics make robotic haircutting a distinctive and challenging robotics problem rather than a straightforward extension of existing grooming automation.

Although haircutting presents unique constraints, it also shares a structural similarity with other coverage-oriented robotic tasks. A useful analogy can be drawn between robotic lawn mowing and robotic haircutting: in both cases, the robot must systematically traverse a target domain and remove material to a prescribed level. This analogy, illustrated in Figure 1, suggests that robotic haircutting can be interpreted as a coverage-oriented autonomy problem and therefore may benefit from ideas developed in mature robotic domains. Compared with robotic lawn mowing, haircutting operates on a human head with sensitive anatomy and strict safety requirements. The manipulated medium



Figure 1. Analogy between robotic lawn mowing and robotic haircutting, highlighting the natural extension of mobile robot techniques from grass mowing to hair trimming.

is also highly variable across users in texture, density, curl pattern, and moisture (Da Costa, 2007; Harrison & Sinclair, 2003). Moreover, the desired outcome is not merely full coverage, but a personalised hairstyle achieved under aesthetic and comfort constraints. These differences mean that haircutting cannot be addressed by directly transplanting existing coverage solutions.

A core difficulty in this emerging area is that research on robotic haircutting remains sparse and fragmented. Existing studies are distributed across a small number of prototype systems, adjacent grooming tasks, perception and hair modelling methods, end-effector design, and learning-based hairstyle generation, but these directions are rarely organised within a unified autonomy framework. As a result, the field still lacks a systematic perspective that explains how perception, planning, and execution are coupled in robotic haircutting. This fragmentation makes it difficult to compare methods, identify core technical bottlenecks, and formulate future research directions in a coherent way.

To address this gap, this paper examines robotic haircutting from the perspective of mobile robotics. In mobile robotics, autonomy is commonly understood through three tightly connected core problems: localisation, planning, and control. This analytical structure has supported progress across a wide range of systems, from autonomous vehicles to household service robots. As illustrated in Figure 2, we argue that the same perspective provides a principled entry point for understanding robotic haircutting as a constrained coverage task. Under this perspective, localisation concerns estimating the spatial relationship between the

cutting tool and the human head; coverage planning concerns generating trimming trajectories that progressively transform the initial hair volume into a target hairstyle; and execution concerns the safe, compliant, and efficient realisation of the planned motion under disturbances such as head movement, hair variability, and tool-surface interaction. Building on this perspective, the main contribution of this work is not merely to list existing studies, but to provide a structured review framework for robotic haircutting. Specifically, the contributions are threefold:

- systematically organises automatic robotic haircutting in terms of the three canonical problems of autonomy, including localisation, coverage planning, and control;
- introduces a volumetric modelling perspective that formulates haircutting as a constrained multi-pass coverage process, thereby linking hairstyle transformation to planning and evaluation;
- discusses broader system-level issues, including learning-driven frameworks, safety considerations, standardisation, and future deployment challenges, in order to outline a more complete research agenda for this emerging field.

The remainder of this paper is organised as follows. Section 2 reviews existing work on robotic haircutting and related areas, and identifies the main gaps that motivate a unified autonomy-oriented framework. Section 3 examines localisation in haircutting robots, covering sensing foundations, representative estimation strategies, and evaluation metrics. Section 4

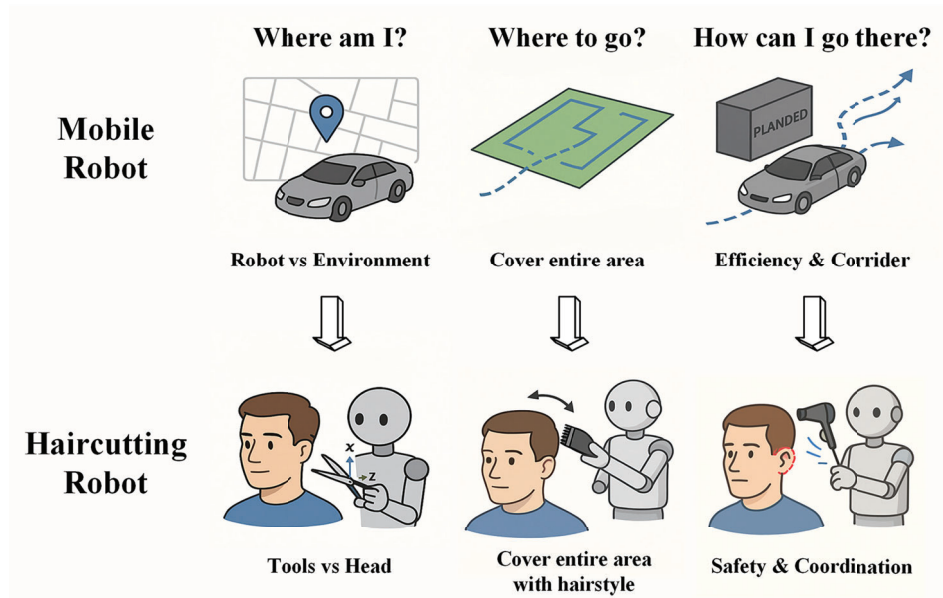


Figure 2. Conceptual mapping of the three core problems in mobile robotics, including localisation, coverage planning, and execution, to the domain of robotic haircutting.

reformulates haircutting as a coverage planning problem by introducing volumetric modelling, recursive trimming, and safety-aware constraints. Section 5 discusses execution and control, with emphasis on actuator design, control strategies, and performance metrics for safe and efficient task realisation. Section 6 then considers learning-driven frameworks and insights from autonomous systems that may support adaptation and integration in robotic haircutting. Finally, Section 7 concludes the paper by summarising the unified perspective developed in this review.

2. Related work

Direct research on robotic haircutting remains limited and has not yet evolved into a mature subfield with clearly defined research branches. Existing work is scattered across early prototype systems, contact-rich hair manipulation, perception and hair modelling, and a small number of survey and perspective studies. This fragmentation makes it necessary to clarify the current scope of robotic haircutting research, distinguish the main lines of work, and identify the key knowledge gaps that motivate this review.

Early concepts appeared in patents for automatic haircutting machines (Gronier, 1966). Later, researchers explored camera-equipped robotic support for self-haircuts and introduced a multi-camera

configuration to capture around-head visual information for haircut assistance (Kyosuke et al., 2014). More recently, DIY prototypes have emerged, including Shane Wighton's depth-camera-guided haircutting arm (Stuff Made Here, 2021) and a UR5e-based robot (Manish Sinha, 2020). These systems demonstrated the basic feasibility of robot-assisted haircutting, but generally lacked systematic autonomy. In contrast, commercial efforts have more often focused on head-care and grooming-related functions than on haircutting itself. Representative examples include Panasonic's hair-washing robot (Panasonic, 2011), whereas haircutting-oriented commercial or patented devices have remained comparatively limited (Aldabbah, 2023). Academic research has since explored more task-relevant directions for near-head manipulation, including the University of Tokyo's strand-oriented styling robot (Kim et al., 2025), MIT's RoboWig with tactile and visual sensing (Hughes et al., 2021), and Moe-Hair's soft end-effectors for compliant manipulation (Yoo et al., 2024; Yoo et al., 2025). Related work on robotic combing and brushing further highlights the importance of safe, contact-rich interaction near the head (Dennler et al., 2021). On the perception side, hair modelling methods such as HairNet (Zhou et al., 2018), AutoHair (Chai et al., 2016), Neural Haircut (Sklyarova et al., 2023), and Gaussian Haircut (Zakharov et al., 2024) reconstruct 3D geometry from monocular images or videos and provide

useful priors for hair representation and style modelling. However, these studies mainly address visual reconstruction and appearance modelling rather than the full robotic haircutting pipeline.

Taken together, the literature reveals three main patterns. First, direct robotic haircutting studies remain scarce and are mostly limited to proof-of-concept systems, end-effector designs, or isolated demonstrations. Second, much of the existing work concentrates on vision-related topics, such as hair reconstruction, hairstyle representation, and generative modelling, but does not address how a robot should localise the cutting tool, plan safe trimming trajectories, and execute them robustly near the human head. Third, the field still lacks a unified task-level formulation that explicitly connects perception, planning, execution, and evaluation. Consequently, robotic haircutting is seldom framed as a closed-loop problem with explicit hair-state estimation, goal specification, constrained trajectory generation, and disturbance-aware execution, making systematic comparison and evaluation difficult.

Recent studies have approached robotic haircutting from different perspectives. Safety, for example, has been identified as a key concern in robotic haircutting (Huang et al., 2025). More general review and perspective articles have examined the topic in terms of systems, methods, technical challenges, hair modelling, planning, control, industrial prospects, and social value. One review summarised robotic haircutting systems, methods, technical challenges, and hair modelling insights (Khan & Li, 2026b), while a more recent CNC-inspired survey framed robotic hair cutting as a form of precision personal care automation, emphasising system architectures, sensing modalities, hair modelling, toolpath planning, and control (Khan & Li, 2026a). Li provided a broader account spanning system design, perception, planning, execution, and industrial prospects (Li, 2025a), whereas another perspective study discussed the topic in terms of social value, market potential, and future directions (Li, 2025b). Collectively, these works have improved the visibility and organisation of the area. However, they have not yet consolidated the field around a unified autonomy-oriented formulation that explicitly links localisation, coverage planning, execution, and evaluation.

Against this background, the contribution of the present review is not simply to expand the list of

reported studies, but to reorganise robotic haircutting through the three tightly coupled autonomy problems of localisation, coverage planning, and execution. Rather than claiming that the existing literature has already matured along these lines, this review adopts them as an analytical framework for interpreting fragmented studies, identifying unresolved technical gaps, and clarifying a more coherent research agenda for the field.

3. Where am i?

Localisation is essential for haircutting robots, as it supports trajectory planning and safe interaction. In mobile robots, localisation typically refers to estimating the robot's pose within a global environment, whereas in haircutting robots the focus shifts to estimating the relative clipper-head pose, with the head itself subject to motion and deformation. This estimation relies on multimodal sensing, which provides geometric context, motion cues, and contact safety. The achievable accuracy and robustness depend on how these complementary signals are combined. From the perspective of mobile robotics, localisation can be organised into three levels: *global localization* for consistent reference frames, *relative localization* for real-time tracking, and *integrated fusion* for robust estimation. This section reviews sensing foundations, representative localisation strategies, and evaluation metrics, highlighting how mobile robotics methods can be adapted to haircutting scenarios.

3.1. Sensing foundations for localization

To enable reliable relative clipper-head pose estimation, sensing forms the foundation of all localisation strategies. In mobile robotics, multimodal perception underpins localisation by combining complementary sensing modalities. Haircutting robots can inherit this principle and adapt it to the specific challenge of estimating tool-head pose under close human-robot interaction. Figure 3 illustrates representative sensing modalities adapted from mobile robots, while Table 1 provides a systematic overview of sensor categories, typical examples, and their roles.

Beyond individual functions, the main challenge in haircutting robots lies in how these modalities interact. Long-range sensing, such as ultrasonic and LiDAR, offers geometric awareness and collision avoidance but

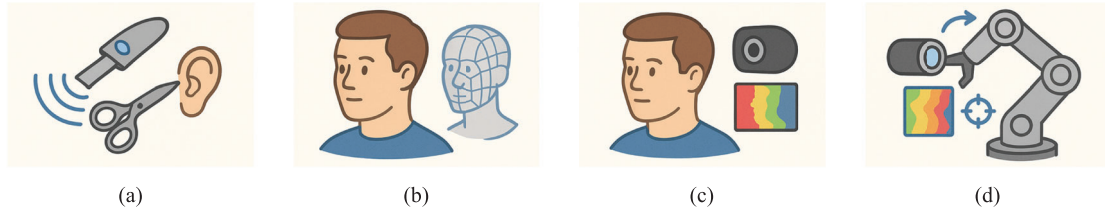


Figure 3. Representative sensing modalities adapted from mobile robotics and applied to haircutting robots. (a) Ultrasonic sensing for collision avoidance. (b) LiDAR-based 3D head reconstruction. (c) RGB-D cameras for head modelling and tracking. (d) Joint encoders for precise tool localisation.

Table 1. Sensor categories and their localisation roles in haircutting robots.

Category	Representative sensors	Localisation role in haircutting robots
Range sensing	Ultrasonic, infrared, capacitive, mmWave radar	Near-field obstacle detection, scalp clearance monitoring, and basic safety protection
Depth sensing	Structured light, ToF cameras, 2D/3D LiDAR	Head surface reconstruction, geometric initialisation, and coarse spatial perception
Vision sensing	RGB cameras, stereo vision, RGB-D cameras, event cameras	Head pose estimation, hairstyle observation, and robust target tracking under motion
Proprioceptive sensing	Joint encoders, IMUs, joint torque sensors	End-effector state estimation, kinematic consistency maintenance, and tool localisation
Contact sensing	Force/torque sensors, tactile arrays, electronic skin	Contact-state feedback, compliant positioning, and safe hair-scalp interaction monitoring

suffers from occlusion and reflection issues on hair surfaces. Vision-based sensors capture rich geometry and texture cues but are sensitive to lighting and motion. Proprioceptive and tactile sensors provide stable local feedback and immediate safety guarantees, yet lack a global reference. No single modality is sufficient in isolation; effective localisation arises from their integration. Another important factor is temporal resolution. High-frequency proprioceptive and tactile sensors enable rapid reaction to sudden head motion, whereas lower-frequency global or depth sensors maintain consistency over longer horizons. Balancing these complementary characteristics is essential for reliable trimming.

The integration of these heterogeneous signals through sensor fusion further improves robustness against individual sensor failures and mitigates uncertainty. By combining global geometric awareness with fine-grained local feedback, haircutting robots can achieve the real-time accuracy required for safe and efficient trimming.

3.2. Localisation strategies

While sensing provides raw observations, localisation strategies determine how these signals are used to estimate the clipper–head pose. Following the perspective of mobile robotics, three levels can be distinguished: *global localization*, *relative localization*, and *integrated fusion*. Table 2 summarises representative methods,

their roles in mobile robots and haircutting robots, and current limitations.

Global localisation establishes an absolute pose in a consistent coordinate frame, providing a stable geometric foundation. Outside-in vision using fixed cameras or RGB-D sensors offers reliable initialisation (Xu et al., 2023), either through fiducial markers or markerless head modelling based on 3D morphable models (Banz & Vetter, 2003) or template alignment (Caporaso et al., 2010). Alternatively, kinematic estimation from robotic manipulators provides deterministic pose computation once hand–eye calibration is available (Horaud & Dornaika, 1995; Ma et al., 2025). External tracking systems such as motion capture, magnetic tracking, or UWB beacons can also be employed, but all global approaches depend on careful calibration and are sensitive to head motion.

Relative localisation focuses on incremental motion tracking and provides responsiveness without external infrastructure. Visual odometry (Fraundorfer & Scaramuzza, 2012), optical flow (Guizilini et al., 2022), and visual–inertial odometry are lightweight solutions but suffer from occlusion and texture homogeneity in dense hair regions (Onkarappa & Sappa, 2014). Proprioceptive sensing with IMUs (Alatise & Hancke, 2017) and encoders enables continuous updates, though drift accumulates over time (Sugino et al., 2024). Contact and proximity sensing

Table 2. Mapping of localisation strategies from mobile robots to haircutting robots.

Category	Representative methods	Role in mobile robots	Mapped role in haircutting robots	Main limitation
Global localisation	Outside-in vision, external tracking, absolute kinematic registration	Absolute pose estimation in a shared reference frame for initialisation and global consistency	Absolute clipper-head registration for trimming initialisation and reference alignment	Requires external infrastructure and calibration; sensitive to head motion and occlusion
Relative localisation	Visual odometry, visual-inertial odometry, encoder/IMU odometry, optical flow, force/tactile feedback	Short-term motion tracking in a local frame for continuous state update	Continuous tracking of clipper motion relative to the moving head during trimming	Drift accumulation; local accuracy degrades without global correction
Integrated fusion	EKF/UKF/PF, factor-graph optimisation, sliding-window estimation, learning-based fusion	Multi-sensor fusion for robust localisation under uncertainty and dynamic disturbances	Fusion of vision, proprioception, and contact feedback for robust close-range trimming	Higher computational and synchronisation cost; performance depends on calibration quality

supply direct safety feedback by constraining penetration depth, which is critical in human-robot interaction (Arshad et al., 2023; Kim et al., 2015; Maiolino et al., 2013), but they lack a global reference.

Integrated fusion combines global stability with local responsiveness, improving both accuracy and robustness. Filtering-based methods such as Kalman filters (Roumeliotis & Bekey, 2000), extended Kalman filters (Ahmad & Namerikawa, 2013), unscented Kalman filters (Naab & Zheng, 2022), and particle filters (Raja et al., 2021) handle sensor noise in real time. Optimisation-based approaches such as factor graph optimisation and sliding-window estimators (Li et al., 2018; Liu et al., 2023; Zhang et al., 2021) achieve long-horizon consistency. More advanced frameworks employ tight coupling of modalities, such as vision-IMU-contact fusion (Deng et al., 2025), often requiring delay compensation and synchronisation. Recently, learning-based fusion has emerged, using deep networks or graph neural models to align heterogeneous sensor streams. Practical requirements, such as online calibration and outlier rejection, are essential to ensure safe and real-time operation. Through such multi-sensor integration, haircutting robots can achieve the robustness required for trimming in dynamic, unstructured environments.

3.3. Evaluation metrics in localization

From a localisation perspective, the core problem in haircutting robots is to estimate the clipper-head pose reliably despite human motion and complex hair geometry. In mobile robotics, evaluation typically emphasises *geometric accuracy* and *robustness*, as tasks are defined with respect to global maps or navigation goals. By contrast, haircutting robots operate in close physical proximity to humans, where localisation

errors may directly cause unsafe contact. Therefore, evaluation must additionally emphasise *safety-related metrics*, alongside latency requirements to ensure real-time adaptation. Evaluation can be structured into four categories:

- **Accuracy:** absolute trajectory error (ATE) and relative pose error (RPE) quantify global consistency and short-term drift, respectively, while surface registration error measures the alignment between the estimated head model and ground truth scans.
- **Robustness:** performance under occlusion, visual degradation, or partial sensor failure reflects the resilience of the localisation pipeline.
- **Latency and update rate:** estimation delay and refresh frequency indicate whether estimates can support real-time trimming without accumulating risk.
- **Safety-related metrics:** minimum scalp distance violation rate, maximum contact force exceedance, and false penetration events capture whether localisation errors translate into unsafe tool-head interactions.

These evaluation dimensions provide a comprehensive basis for comparing localisation approaches and identifying gaps when transferring methods from mobile robotics to haircutting tasks. Together with sensing foundations and algorithmic strategies, they complete the localisation perspective of the unified framework.

4. Where should i go?

In the task of autonomous haircutting, determining how the clipper should traverse the scalp to achieve a specific hairstyle constitutes a core challenge. From

the perspective of mobile robotics, this problem can be framed as a *Coverage Path Planning* (CPP) task, which aims to compute a trajectory that ensures the complete and efficient coverage of a target area. Similar to floor cleaning or lawn mowing robots, haircutting systems must plan the motion of their end-effector to cover a three-dimensional domain. However, unlike traditional CPP tasks that often focus on 2D ground surfaces, haircutting introduces several distinctive complexities:

- **Multi-pass trimming:** Due to the limited thickness that the clipper can remove in a single pass, the desired hair volume cannot be achieved through one-time motion. Instead, the system must perform a sequence of overlapping trimming passes, each of which must be precisely aligned and depth-regulated to progressively shape the final result.
- **Shaped outcome requirement:** Unlike generic area coverage tasks, where the objective is to visit all regions equally, haircutting demands conformity to a predefined three-dimensional target shape. This necessitates that the planned trajectories not only achieve full surface coverage but also sculpt the hair volume to meet aesthetic and functional design specifications.
- **Safety constraints:** The robot must avoid contact with sensitive anatomical regions such as the ears, eyes, and face. This requires the explicit integration of spatial exclusion zones into the coverage planning algorithm to ensure safe tool operation throughout the trimming process.

4.1. Geometric modeling of clipper motion

It is necessary to quantify the 3D region affected by the clipper as it moves along a planned trajectory, because this region determines both the effectiveness and safety of the trimming operation. This region, referred to as the swept volume, represents the spatial envelope traced by the clipper and serves as a geometric foundation for evaluating trimming completeness and enforcing collision constraints.

Definition 4.1: (Instantaneous occupied region of the clipper).

Let $K \subset \mathbb{R}^3$ denote the geometric model of the clipper in its local coordinate frame. During the i -th

trimming pass, the clipper pose $g_i(t)$ at time $t \in I_i$, where $I_i = [t_i^{\text{start}}, t_i^{\text{end}}] \subset \mathbb{R}$ denotes the time interval of the i -th trimming pass, is described by the rigid-body transformation

$$g_i(t) = (R_i(t), p_i(t)) \in SE(3),$$

where $R_i(t) \in SO(3)$ is the rotation matrix and $p_i(t) \in \mathbb{R}^3$ is the translation vector. The region occupied by the clipper in the global coordinate system is defined as

$$g_i(t) \otimes K := \{R_i(t)u + p_i(t) | u \in K\}. \quad (1)$$

Here, u denotes an arbitrary point in the clipper geometry K expressed in the local coordinate frame, and the symbol $\otimes$ represents the action of a rigid-body transformation on a geometric set, mapping the clipper model K from its local coordinate frame to its occupied region in the global coordinate system.

Equation (1) represents the instantaneous occupied region of the clipper in the global coordinate system. It characterises the tool geometry at time t , rather than the actual cutting or contact region.

Definition 4.2: (Swept volume of the i -th pass).

The swept volume of the i -th trimming pass, denoted by $\mathcal{W}_i$, is defined as the union of all occupied regions over the motion interval:

$$\mathcal{W}_i := \cup_{t \in I_i} g_i(t) \otimes K \subset \mathbb{R}^3. \quad (2)$$

It should be noted that Equation (2) captures the full space traversed by the clipper during the pass, regardless of whether the corresponding material has already been removed. It therefore provides a consistent basis for subsequent modelling of effective trimming, residual hair volume, and coverage optimisation.

4.2. Recursive volume update and trimming outcome

In practice, the swept volume $\mathcal{W}_i$ of a trimming pass does not always correspond to actual hair removal. The clipper may pass through space without cutting, a situation referred to as *air-cutting*. Typical causes include excessive standoff distance due to the guard, blade deactivation, unsuitable orientation, or insufficient contact force. To capture this distinction, we introduce an engagement set $\mathcal{A}_i \subset \mathbb{R}^3$, which represents the subset of space in which cutting is effectively realised during pass i .

Definition 4.3: (Effective removal volume and residual hair volume).

Let $S_0 \subset \mathbb{R}^3$ denote the initial uncut hair volume. For each pass $i = 1, 2, \dots, N$, the effective removal volume E_i and the residual hair volume S_i are defined by

$$\begin{cases} E_i := (\mathcal{W}_i \cap S_{i-1}) \cap \mathcal{A}_i, \\ S_i := S_{i-1} \setminus E_i, \end{cases} \quad (3)$$

where E_i denotes the portion of hair actually removed during pass i , and S_i denotes the residual hair volume afterwards.

This formulation ensures that only engaged regions contribute to trimming, while air-cutting leaves the hair volume unchanged. It also guarantees that each region of hair is removed at most once, thereby providing a consistent foundation for coverage evaluation and multi-pass optimisation. After multiple passes, it is useful to quantify the total amount of hair removed, independent of the order or redundancy of coverage.

Definition 4.4: (Total effective removal volume).

The total effective removal volume after N passes, denoted by E_{total} , is defined by

$$E_{\text{total}} := \bigcup_{i=1}^N E_i = S_0 \setminus S_N. \quad (4)$$

This definition provides a consistent basis for evaluating trimming completeness, comparing different coverage strategies, and optimising multi-pass planning. It should be noted that the volumetric representation introduced above mainly serves as an initial abstraction for coverage planning, rather than implying that hair is treated as a fully static geometric object. In practical execution, local hair shape and trimming outcome may vary during contact. The proposed framework can therefore be further combined with closed-loop perception, for example by updating hair geometry or residual volume estimates online and applying local path correction or iterative replanning when necessary, to improve adaptability and robustness in real grooming scenarios.

4.3. Coverage constraints

Considering the main factors involved in haircutting coverage planning, especially safety and hairstyle requirements, we define two fundamental constraints: *trimming depth* and *collision avoidance*. These constraints determine how much hair can be removed in

each pass and where the clipper can safely move. Each constraint is first defined formally and then illustrated through its special cases.

4.3.1. Trimming depth constraint

Definition 4.5: (Trimming depth constraint).

For each pass, the maximum amount of hair removed is limited by

$$\begin{aligned} \text{depth}(x, S_{i-1}) &:= \inf_{y \in \partial S_{i-1}} \|x - y\| \\ &\leq d_{\max}, \quad \forall x \in E_i, \end{aligned} \quad (5)$$

where y denotes an arbitrary point on the boundary of the residual hair volume S_{i-1} , and $d_{\max} > 0$ specifies the maximum allowable removal thickness relative to the pre-pass boundary ∂S_{i-1} .

Here, $\text{depth}(x, S_{i-1})$ serves as a geometric surrogate for the local removal thickness during pass i . By construction, each trimming pass removes only a thin layer of hair, thereby preventing large, irreversible cuts and allowing gradual refinement of the hairstyle. Accordingly, this constraint can be discussed under two special cases:

- **Single-pass trimming** ($d_{\max} = \infty$): If trimming depth is unconstrained, the clipper is assumed to remove the desired hair volume in one pass. The task then reduces to a simplified single-pass volumetric coverage problem, analogous to classical CPP tasks such as floor cleaning or lawn mowing, where the objective is complete coverage with minimal overlap and execution time.
- **Multi-pass trimming** ($d_{\max} < \infty$): In realistic haircutting, trimming depth is limited and hair styles require layered shaping. The task must therefore be decomposed into successive passes, typically involving:
- **Coarse trimming**: quickly removes a large portion of hair to approximate the rough silhouette of the target hairstyle;
- **Fine finishing**: refines edges, transitions, and surface smoothness to meet aesthetic requirements.

4.3.2. Collision avoidance constraint

Definition 4.6: (Collision avoidance constraint).

Let $O \subset \mathbb{R}^3$ denote the obstacle volume, including sensitive anatomical structures (e.g. ears, eyes, and

nose) and restricted workspace regions near the scalp. At time $t \in I_i$, the occupied volume of the clipper system is

$$K_{\text{full}}(t) := g_i(t) \otimes K \cup \mathcal{L}_i(t), \quad (6)$$

where $\mathcal{L}_i(t)$ denotes the spatial extent of the supporting mechanical links associated with the current trimming configuration. The non-penetration requirement is

$$K_{\text{full}}(t) \cap O = \emptyset, \forall t \in I_i. \quad (7)$$

This constraint ensures that neither the tool nor its supporting links collide with sensitive regions or re-contact already trimmed surfaces. It can likewise be discussed under idealised and realistic workspace conditions:

- **Collision-free workspace** ($O = \emptyset$): If no obstacle regions are considered, the constraint vanishes and the task reduces to volumetric coverage subject only to depth limits. This resembles CPP in open, obstacle-free environments.
- **Realistic workspace** ($O \neq \emptyset$): In haircutting, sensitive anatomical regions must always be excluded. Planning must therefore incorporate explicit exclusion zones, balancing trimming completeness with safety. This parallels mobile robots navigating cluttered environments, where obstacle-aware coverage is essential.

4.4. Coverage planning formulation

Given trimming depth and collision avoidance as fundamental constraints, the haircutting coverage planning problem can be formulated as follows.

4.4.1. Feasible formulation

Definition 4.7: (Feasible haircutting coverage planning problem).

The planning task is to find a sequence of trajectories $\{g_i(t)\}_{i=1}^N$ such that the final residual hair satisfies

$$T^{-\varepsilon} \subseteq S_N \subseteq T^{+\varepsilon}, \quad (8)$$

while satisfying the trimming depth and collision avoidance constraints. Here, $T \subset \mathbb{R}^3$ is the target hairstyle, and $T^{+\varepsilon}$ and $T^{-\varepsilon}$ denote its outer and inner offset sets with tolerance margin ε .

A feasible solution guarantees that the final residual hair volume lies within an ε -tolerance band around the target hairstyle, while all tool motions remain safe and admissible.

4.4.2. Optimal formulation

Beyond feasibility, the planner may further optimise performance criteria. In haircutting, two objectives are particularly relevant:

- **Time-optimal execution:** complete the trimming task in the shortest possible time, reducing delays caused by velocity limits, direction changes, or reconfiguration;
- **Energy-efficient execution:** minimise energy consumption, which may be approximated by cumulative actuation effort, control input, or total path length, subject to feasibility.

Accordingly, a general optimal haircutting coverage planning problem can be expressed as

$$\min_{\{g_i(t)\}_{i=1}^N} (J_{\text{time}}, J_{\text{energy}}) \quad (9)$$

subject to the recursive volume update, trimming depth constraint, collision avoidance constraint, and terminal tolerance-band condition defined above. Here, J_{time} denotes the total execution time, while J_{energy} represents the overall energy cost under the adopted system model.

In practice, these objectives are not always consistent. A path with shorter geometric length may still require frequent stops, reorientation, or sharp turns, thereby increasing execution time, whereas a smoother trajectory may reduce execution time at the cost of greater actuation effort. Haircutting coverage planning is therefore naturally posed as a multi-objective optimisation problem, in which Pareto-optimal solutions balance time efficiency and energy consumption.

5. How can i go there?

The final component of autonomous haircutting is *execution*, in which planned trajectories must be realised safely, efficiently, and with the desired trimming quality. This process relies critically on the actuation system, which determines how the robot generates motion and interacts with the human head.

Table 3. Representative actuators and their potential function in haircutting robots.

Category	Representative actuators	Potential function in haircutting robots
Motion actuation for positioning	DC/BLDC motors, Servo motors, Stepper motors, Linear actuators, Pneumatic actuators	Driving robotic arm joints, end-effector orientation, vertical lift, and compliant positioning
End-effector actuation	Vibration/oscillating motors (clipper blades), Rotary servo motors (scissor actuation), Micro linear actuators (guard adjustment), Solenoids	Executing actual trimming, scissor opening/closing, adjusting cutting length, and fast tool release
Adaptive and compliant actuation	Series elastic actuators (SEA), Variable stiffness actuators (VSA), Magnetorheological clutches/dampers	Ensuring safe contact with scalp, force control, compliance tuning, and smooth interaction
Auxiliary actuation	Tool changer actuators, Combing motors, Vacuum motors, Blower/fan actuators, Safety release actuators	Supporting functions such as tool switching, active hair alignment, cut hair collection, airflow assistance, and emergency tool retraction
Human–robot interaction actuation	Haptic feedback actuators (vibration motors), Audio output actuators (miniature speakers)	Providing user alerts, touch cues, and voice-based interaction during trimming

5.1. Actuators

Execution in haircutting robots ultimately depends on actuators, which provide the physical means to move the clipper, perform trimming, and adapt safely to human interaction. Compared with mobile robots, haircutting requires not only accurate motion of a robotic arm but also specialised actuation at the end-effector for hair removal, as well as auxiliary mechanisms for interaction and task support. Typical haircutting-specific actuators include vibration motors embedded in clippers for cutting, combing and brushing mechanisms for hair alignment, and miniature fans or suction units for hair debris management. These are complemented by actuation technologies inherited from broader robotics domains, such as series elastic actuators (SEA) (Robinson, 2000) and variable stiffness actuators (VSA) (Wolf et al., 2016), which enable compliant interaction and enhance safety during contact with the scalp. The integration of specialised and general-purpose actuators supports both the functionality of hair processing and the reliability of motion execution.

Actuator design directly influences safety and personalisation. Compliance and force-limiting mechanisms mitigate the risk of injury during close contact, while modular or adjustable actuators allow adaptation to diverse head geometries and hairstyle preferences. Table 3 summarises representative actuator types and their corresponding roles in haircutting robots. The coordinated use of these actuators enables precise trajectory execution, safe contact with the scalp, and user-friendly interaction, thereby forming the foundation for the control methods discussed in the following sections.

5.2. Solution strategies

Building upon the objectives and constraints discussed above, the next step is to determine concrete strategies for haircutting coverage planning. Existing approaches can be broadly grouped into two complementary categories: *traditional CPP algorithms*, originally developed in mobile robotics, and *learning-driven approaches*, which have emerged from data-centric AI. Table 4 provides a structured comparison of representative methods, illustrating how canonical strategies from mobile robotics can be reinterpreted for haircutting tasks.

5.2.1. Traditional coverage path planning algorithms

In mobile robotics, traditional CPP algorithms provide strong theoretical foundations and deterministic guarantees for workspace coverage. For instance, boustrophedon (lawnmower) coverage and spiral coverage are widely adopted in structured environments such as warehouses or agricultural fields. Their appeal lies in their systematic sweeping motions, which minimise turning overhead and ensure coverage completeness. When transferred to haircutting, these strategies correspond to dividing the scalp into local patches and executing layered trimming passes to cover curved regions. Path smoothing or curvature-aware variants are often necessary to prevent uneven cutting in high-curvature areas such as the crown. Similarly, decomposition-based ideas, which have been widely studied in robotics for reducing complex workspaces into smaller cells, naturally extend to partitioning the scalp into subregions. However, unlike floors or fields, hair is deformable, and repeated multi-pass trimming may be required to achieve uniform results.

Table 4. Representative execution strategies for coverage planning and their applicability to haircutting robots.

Planning paradigm	Category	Representative methods	Key characteristics	Applicability to haircutting
Traditional CPP	Pattern-based	Boustrophedon coverage (Bähnmann et al., 2021) Spiral coverage (Cabreira et al., 2018)	Alternating sweeping motions, efficient for structured areas Outward/inward radial sweeping, suitable for circular regions Exact workspace partitioning into simple cells	Applied to scalp partitioning into local patches, enabling systematic layered trimming
	Decomposition-based	Boustrophedon decomposition (Choset & Pignon, 1998)		Supports segmentation of scalp surface into manageable sub-regions for coverage planning
	Graph-based	Grid/quadtrees subdivision (Fu et al., 2021; Han et al., 2022) Chinese Postman problem (Thimbleby, 2003)	Hierarchical partitioning into uniform or adaptive cells Ensures traversal of all edges with minimum cost Connects regions with minimal cumulative distance	Facilitates scalp-wide coverage planning with global path optimisation
	Sampling-based	Minimum Spanning Tree (Gabriely & Rimon, 2001) Probabilistic Roadmap (PRM) (Kavraki et al., 1998) Rapidly-exploring Random Tree (RRT) (Umari & Mukhopadhyay, 2017)	Builds graph of feasible connections in configuration space Efficient exploration of high-dimensional spaces Gradient-free and population-based metaheuristics enabling trade-offs among multiple criteria	Connects local scalp patches or handles constrained regions with complex geometry
	Multi-objective optimisation	Heuristic-based methods: Single-agent (BAS(Khan et al., 2022)); Swarm-based (GA (Lamini et al., 2018), PSO(Lin et al., 2023), ESOA (Huang et al., 2024)) Pareto-based methods (Zhao & Zhu, 2020)		Balances efficiency, energy consumption, safety, and aesthetic quality during trimming
Learning-driven	Imitation learning	Dynamic neural networks (Li et al., 2020; C. Hua et al., 2026) Behaviour cloning (Jang et al., 2022)	Evolutionary multi-objective optimisation maintaining Pareto fronts and solution diversity Neural dynamics optimisation methods with finite-time convergence under constraints Directly learns from expert demonstrations	Captures stylistic features of professional barbers and transfers them to robotic execution
	Reinforcement / hybrid learning	Demonstration-based deep models (Kim et al., 2023; Zare et al., 2024) Reinforcement learning (RL) (Han et al., 2023; Ibarz et al., 2021; Zhu & Zhang, 2021) RL + imitation / optimisation-enhanced RL (Hua et al., 2021)	Learns motion policies with stylistic consistency Policy optimisation via trial-and-error with rewards Hybrid paradigms combining demonstrations with optimisation	Adapts trimming trajectories to user preferences and dynamic hair properties through simulation-trained policies

Beyond purely geometric methods, multi-objective optimisation frameworks have also been employed in coverage problems. In mobile robotics, such methods are used to trade off path length, energy efficiency, and obstacle avoidance. In haircutting, the objectives further extend to trimming smoothness, hairstyle symmetry, and user comfort. This often motivates the use of metaheuristic approaches to search for balanced solutions across competing objectives. While these methods can incorporate aesthetic and safety constraints more explicitly than purely geometric coverage strategies, they are limited by convergence speed and the lack of real-time guarantees, especially in interactive settings involving human users.

In summary, traditional methods contribute determinism and efficiency, making them valuable as a structural backbone. However, their limitations in handling scalp-geometry variability, hair dynamics, and multi-objective trade-offs highlight the need for augmentation by adaptive or data-driven modules.

5.2.2. Learning-driven strategies

Learning-driven approaches aim to capture personalisation, stylistic variation, and the inherent uncertainties of hair dynamics, which are difficult to encode in purely geometric planning. Reinforcement learning has been widely applied in navigation and manipulation domains (Han et al., 2023; Ibarz et al., 2021; Zhu & Zhang, 2021), where agents optimise strategies through trial and error in simulation. In haircutting, similar formulations allow the robot to balance time, energy, and trimming quality via carefully designed reward functions. This enables adaptation to diverse head geometries and motion conditions that are difficult to handle with deterministic CPP alone.

Imitation learning provides another promising paradigm by transferring expert demonstrations into robot policies. Studies in manipulation and motion generation (Jang et al., 2022; Kim et al., 2023; Zare et al., 2024) have shown that imitation learning preserves stylistic consistency and natural motion patterns. For haircutting, this translates into learning the nuanced strategies of professional barbers directly from demonstrations, which can serve as effective initialisation for further reinforcement learning.

The main challenge lies in data availability and safety, since large-scale demonstrations or simulations are required, and learned policies must generalise without compromising user safety. Nevertheless,

learning-driven approaches complement traditional CPP by enabling adaptation to individual users and aesthetic preferences, thereby offering the flexibility needed for real-world deployment.

5.3. Evaluation metrics for execution

From the execution perspective, evaluation focuses on whether planned trajectories can be realised smoothly, safely, and efficiently by the haircutting robot. Unlike localisation, which emphasises pose estimation accuracy, execution metrics emphasise trajectory tracking quality, interaction safety, and the overall efficiency of the trimming process. Representative evaluation categories include:

- **Trajectory tracking performance:** root-mean-square tracking error, maximum deviation, and trajectory smoothness indicate how faithfully the robot follows planned paths while maintaining user comfort.
- **Contact safety:** peak contact force, force overshoot ratio, and scalp contact violation rate capture whether motion execution remains within safe limits during human–robot interaction.
- **Temporal efficiency:** execution time per pass, idle-time ratio, and replanning overhead quantify the time cost of trajectory realisation.
- **Energy efficiency:** cumulative joint power, actuation effort, and energy consumed per unit volume of hair removed measure the energy cost of execution.
- **System resilience:** recovery success rate under disturbances, e-stop response time, and task completion rate under real-time adaptation reflect robustness to unexpected head motion or system failures.

Together, these metrics provide a quantitative basis for comparing different execution strategies, balancing execution efficiency with safety, and guiding the design of hybrid control frameworks in haircutting robots.

6. Broader perspectives and future directions for robotic haircutting

Figure 4 Illustration of a VLA framework for haircutting robots. Cameras track AprilTags to ensure robust localisation, while an IMU mounted on the end effector captures motion dynamics. An LLM-based control

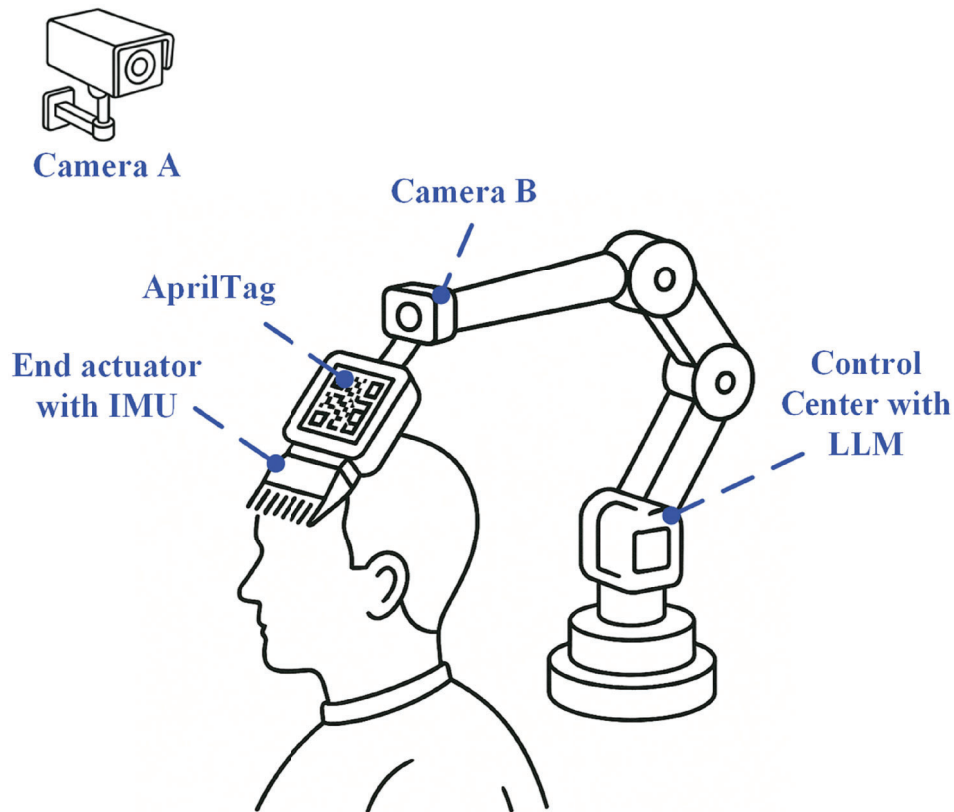


Figure 4. Illustration of a VLA framework for haircutting robots.

centre then fuses sensory inputs with language instructions to generate trimming commands for safe and personalised execution.

The preceding sections have examined the core problems of robotic haircutting from the perspectives of localisation, coverage planning, and execution control. However, advancing robotic haircutting from a conceptual framework to a practically deployable system requires more than refining these individual modules. It also calls for broader technical insights, system-level considerations, and discussion of deployment-related issues. Accordingly, this section extends the discussion by drawing insights from emerging AI technologies and mature autonomous systems, while also addressing key issues related to safety, reliability, standardisation, and future development. In this way, the section aims to provide a broader perspective on the challenges and directions involved in moving robotic haircutting toward real-world application.

6.1. Insights from emerging AI technologies

Learning-based methods have become increasingly important in autonomous systems because they

enhance adaptability in complex and uncertain environments. For robotic haircutting, they are especially useful for handling variations in head geometry, partial occlusion, contact uncertainty, and user-specific preferences.

Deep learning (DL) provides the most direct inspiration for robotic haircutting. Following Zhang et al. (2018), DL can be interpreted as a process that not only learns relations among variables, but also learns the knowledge that governs these relations and the knowledge that makes these relations meaningful. Therefore, it can be viewed as a hierarchical representation-learning framework whose main strength lies in the progressive extraction of task-relevant features from raw sensory data (LeCun et al., 2015). This is particularly important in haircutting, where visual, force, and motion signals are often affected by noise, coupling, and partial occlusion. These signals must be effectively represented before they can support reliable perception of head and tool states, contact state estimation, and stable control. On this basis, DL can also support interaction dynamics modelling and residual compensation for control errors under uncertainty. Existing studies have shown that uncertainty-aware prediction

and robust policy training can reduce risk in close human–robot interaction. These ideas are therefore directly relevant to improving the safety and stability of robotic haircutting (Muhammad et al., 2020). However, an important limitation of DL is the lack of transparency in the internal decision-making process of learned models. A model may produce effective outputs, while the basis for those decisions remains difficult to interpret directly (Wu et al., 2016). For this reason, in near-human and safety-critical robotic haircutting, DL should not replace existing methods entirely. A more appropriate direction is to integrate it with the model-driven frameworks discussed in this work, including localisation, planning, and execution. In this way, learning capability can be embedded into an interpretable and verifiable system structure, leading to a safer and more auditable hybrid system.

Large language models (LLMs) provide a different type of inspiration. Their main contribution lies in high-level task understanding. Robotic haircutting is inherently personalised, and user requirements are often expressed in natural language, such as ‘shorten the sides by 2 cm’ or ‘avoid trimming near the scar’. This suggests that LLMs may help translate user intent and local constraints into structured parameters for planning and execution (Cui et al., 2024). Existing studies have explored LLMs in other autonomous systems for reasoning, memory, and high-level decision support (Fu et al., 2024). In robotic haircutting, LLMs are therefore better viewed as semantic interfaces between user expression and robotic behaviour.

Nevertheless, this direction remains forward-looking rather than a mature solution for haircutting applications. Vision-language-action (VLA) frameworks suggest a more unified multimodal paradigm. These methods aim to directly connect visual perception, language understanding, and action generation (Ma et al., 2024). Recent systems, such as OpenVLA and RT-2, have demonstrated the potential for generalisation across multi-task robotic settings (Kim et al., 2024; Zitkovich et al., 2023). This paradigm is attractive for robotic haircutting because the task naturally depends on multimodal information. Visual input is required to interpret the spatial relationship between the head and the tool, while tactile or force feedback may help assess contact conditions. User preferences must also be incorporated through language.

Within such a framework, AprilTags can provide stable anchors for pose estimation (Krogius et al., 2019; Olson, 2011; Wang & Olson, 2016). An end-effector-mounted IMU can provide inertial information to mitigate drift and improve short-term motion consistency. Together, these elements point to a promising direction for future robotic haircutting systems. However, their safety, reliability, and generalisation under real-world close human–robot interaction conditions still require substantial validation.

Overall, these emerging AI technologies outline a clear pathway for robotic haircutting. DL strengthens low-level perception and adaptation. LLMs support high-level understanding of user intent. VLA frameworks point toward unified multimodal action generation. Their role is not to replace model-driven safety mechanisms, but to complement them. Such integration may help move robotic haircutting toward systems that are more robust, more personalised, and more practically deployable.

6.2. Insights from mature autonomous systems

Beyond learning-driven methods, haircutting robots can also benefit from insights drawn from mature autonomous systems. Autonomous driving (AD) contributes established approaches to trajectory generation, safety filtering, and real-time adaptation, while research on unmanned aerial vehicles (UAVs) provides strategies for three-dimensional coverage and visual–inertial fusion. Although originally developed for dynamic and unstructured environments, these methodologies align closely with the execution challenges encountered in robotic haircutting.

In AD, deep neural networks support perception and decision-making, while control barrier functions (CBFs) (Alan et al., 2023) and model predictive control (MPC) (Williams et al., 2018) enforce safety and optimise trajectories under constraints. End-to-end frameworks further demonstrate how multimodal inputs, such as cameras, LiDAR, and radar, can be mapped directly to steering and acceleration commands with feedback adaptation (Grigorescu et al., 2020). An example is shown in Figure 5, which contrasts the end-to-end pipeline in autonomous driving with its counterpart in haircutting. Both systems adopt multimodal sensing, deep neural networks, and feedback loops, but their control targets differ: steering commands in driving versus trimming trajectories

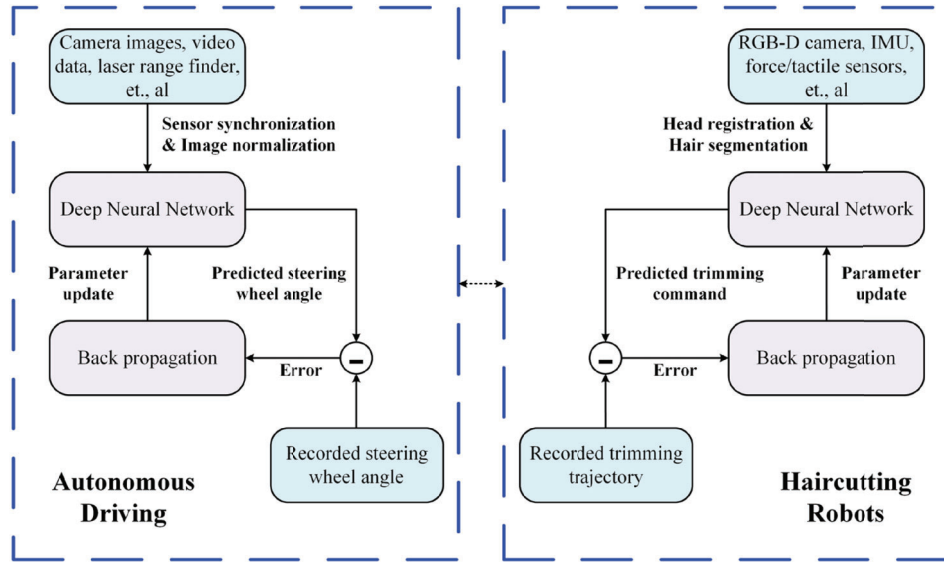


Figure 5. Analogy between autonomous driving and robotic haircutting. Both employ end-to-end pipelines with multimodal sensing, deep neural networks, and feedback adaptation, but differ in control targets: steering commands in driving versus trimming trajectories in haircutting.

in haircutting. This analogy highlights how trajectory smoothing corresponds to clipper path refinement on curved scalp geometry, how velocity constraints map to safe trimming speeds, and how CBF-like mechanisms may enforce exclusion zones around sensitive anatomical regions such as the ears and eyes.

In addition, research on UAVs has produced effective methods for covering complex three-dimensional surfaces, segmenting irregular workspaces, and executing visual servoing in real time (Tsiakas et al., 2024). Such strategies are directly relevant to haircutting robots, where the scalp forms a non-planar and deformable surface, and the trimming process must adapt dynamically to head geometry and motion. Moreover, visual-inertial fusion, which is central to UAV navigation, is directly applicable to tool-head localisation in haircutting systems and may enhance robustness under occlusion or sensor degradation (Zhao et al., 2025).

Future progress in haircutting robots will likely depend on hybrid execution frameworks that combine the stability of model-based control with the adaptability of learning-driven approaches. Integrating these directions into unified system architectures represents a promising path toward robust, safe, and user-centered haircutting systems, thereby supporting the transition from research prototypes to practical deployment.

6.3. Safety, reliability, and standardization considerations

Robotic haircutting is a service-robot application that operates in close proximity to the human body, and direct or indirect human contact is often unavoidable during task execution. Safety should therefore be regarded as a central issue in system design and deployment. For such tasks, safety should not be understood merely as the avoidance of local contact injury. It also involves reliability, robustness, and resilience: the ability to operate consistently under normal conditions, maintain acceptable performance under disturbance and uncertainty, and preserve or recover safe operation under partial damage or abnormal conditions through protection and recovery strategies.

From the perspective of implementation, safety assurance in robotic haircutting can be viewed through two complementary approaches: proactive safety and runtime safety. The former emphasises the incorporation of protective mechanisms during the design stage, for example through layered safety design at the mechanical, sensing, and algorithmic levels (Huang et al., 2025), or through system-level analysis in terms of key functional requirements, collaboration modes, relevant standards, and safety assurance mechanisms (Bi et al., 2021). The latter emphasises the system's capability for protection and recovery under abnormal operating conditions, including obstacle avoidance,

emergency stop, fault detection, safe stop, degraded operation, and recovery based on remaining resources (Zhang et al., 2017). These mechanisms suppress risk propagation under partial failure or functional degradation and help preserve or restore safe behaviour, which also reflects a core aspect of system resilience.

There is still no dedicated safety evaluation framework or standardised set of metrics specifically for robotic haircutting. The design and testing of robotic haircutting systems should therefore be interpreted in light of existing international robot safety standards. ISO 13482:2014 provides general safety requirements for personal care robots and also covers applications involving physical human–robot contact (International Organization for Standardization, 2014), while ISO/TS 15066:2016, although primarily developed for collaborative robots, is also relevant to close-proximity human interaction in robotic haircutting because of its discussion of physical contact risks and contact safety limits (International Organization for Standardization, 2016). Against this background, the metrics adopted in this work, such as contact safety, avoidance of sensitive regions, and failure-related constraints, can be interpreted as task-specific extensions of existing safety evaluation principles for robotic haircutting. Future work should further promote the establishment of dedicated safety metrics, testing protocols, and operational guidelines for robotic haircutting, which will be essential for future research benchmarking, system certification, and broader commercial deployment.

6.4. Future directions

Future research in robotic haircutting should focus not only on increasing autonomy, but also on improving adaptability to realistic grooming conditions. Compared with the current planning framework, which mainly relies on geometric abstraction, practical haircutting involves continuous variations in local hair shape, contact conditions, and trimming outcomes during execution. Accordingly, future work should place greater emphasis on closed-loop perception and online adjustment mechanisms. For example, adaptive visual sensing, force or tactile feedback, and residual-volume estimation can be integrated to update hair geometry and local interaction states in real time, while local path correction or iterative replanning can be introduced whenever the actual trimming

result deviates from the initial plan. In addition, high-fidelity simulation platforms, scalable datasets, and sim-to-real validation will be important for advancing robotic haircutting from proof-of-concept systems toward robust real-world deployment.

Beyond technical development, the real-world deployment of robotic haircutting also involves broader ethical and social considerations. Because such systems may acquire data related to the head, face, and personal appearance, careful attention should be given to data minimisation, informed consent, and the boundaries of data storage and use. Moreover, robotic haircutting affects not only the human body but also personal appearance. As a result, questions of accountability and traceability become especially important in cases of unintended contact, injury, or unsatisfactory haircut outcomes. Public trust is another critical factor, since willingness to accept a robot operating near the head and influencing appearance-related outcomes will directly affect adoption and deployment. Future research should therefore address not only system performance, but also privacy protection, accountability, shared control, and mechanisms for building public trust.

7. Conclusion

This review presented a unified perspective on robotic haircutting by interpreting the task through the autonomy framework of mobile robotics: *Where am I?*, *Where should I go?*, and *How can I go there?* From this perspective, localisation establishes the geometric relationship between the cutting tool and the human head, coverage planning formulates trimming as a constrained volumetric coverage problem, and execution and control address the safe, efficient, and personalised realisation of the planned task. On this basis, the review clarified how sensing, modelling, planning, actuation, and control can be understood as tightly coupled components of haircutting autonomy rather than isolated technical modules.

Rather than proposing a new algorithm, this work aimed to provide a structured interpretation of robotic haircutting from a mobile robotics perspective. The proposed framework helps connect fragmented studies in perception, hair modelling, end-effector design, planning, and safe interaction within a common task-level formulation. In this sense, haircutting robots should not be viewed as isolated innovations, but as

an emerging extension of established robotic autonomy principles into a safety-critical, contact-rich, and appearance-sensitive domain. By positioning the field within a broader robotics context, this review provides a basis for more systematic comparison of existing approaches, clearer identification of unresolved challenges, and more coherent future development. Further advances will depend on continued progress in safe human–robot interaction, realistic modelling, and integrated autonomy for practical robotic haircutting systems.

Acknowledgements

This paper is partially supported by MSCA project No. 101236387.

Author contributions

CRedit: **Zhendai Huang**: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Ameer Tamoor Khan**: Investigation, Resources, Supervision, Writing – review & editing; **Shuai Li**: Conceptualization, Methodology, Supervision, Validation, Writing – review & editing

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The article is focused on conducting a survey and highlighting research opportunities, and as a result, no new data was generated or analyzed. Thus, data sharing is not applicable to this study.

References

- Ahmad, H., & Namerikawa, T. (2013). Extended Kalman filter-based mobile robot localization with intermittent measurements. *Systems Science & Control Engineering*, 1(1), 113–126. <https://doi.org/10.1080/21642583.2013.864249>
- Alan, A., Taylor, A. J., He, C. R., Ames, A. D., & Orosz, G. (2023). Control barrier functions and input-to-state safety with application to automated vehicles. *IEEE Transactions on Control Systems Technology*, 31(6), 2744–2759. <https://doi.org/10.1109/TCST.2023.3286090>
- Alatise, M. B., & Hancke, G. P. (2017). Pose estimation of a mobile robot based on fusion of IMU data and vision data using an extended Kalman filter. *Sensors*, 17(10), 2164. <https://doi.org/10.3390/s17102164>
- Aldabbah, M. (2023). Automatic hair cutter robot (Patent No. WO2023/080812 A1).
- Arshad, A., Saleem, M. M., Tiwana, M. I., ur Rahman, H., Iqbal, S., & Cheung, R. (2023). A high sensitivity and multi-axis fringing electric field based capacitive tactile force sensor for robot assisted surgery. *Sensors and Actuators a: Physical*, 354, 114272. <https://doi.org/10.1016/j.sna.2023.114272>
- Bähnemann, R., Lawrance, N., Chung, J. J., Pantic, M., Siegwart, R., & Nieto, J. (2021). Revisiting boustrophedon coverage path planning as a generalized traveling salesman problem. In *Field and Service Robotics* (pp. 277–290).
- Bi, Z. M., Luo, C., Miao, Z., Zhang, B., Zhang, W.-J., & Wang, L. (2021). Safety assurance mechanisms of collaborative robotic systems in manufacturing. *Robotics and Computer-Integrated Manufacturing*, 67, 102022. <https://doi.org/10.1016/j.rcim.2020.102022>
- Blanz, V., & Vetter, T. (2003). Face recognition based on fitting a 3D morphable model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 25(9), 1063–1074. <https://doi.org/10.1109/TPAMI.2003.1227983>
- Cabreira, T. M., Di Franco, C., Ferreira, P. R., & Buttazzo, G. C. (2018). Energy-aware spiral coverage path planning for uav photogrammetric applications. *IEEE Robotics and Automation Letters*, 3(4), 3662–3668. <https://doi.org/10.1109/LRA.2018.2854967>
- Caporaso, J. G., Bittinger, K., Bushman, F. D., DeSantis, T. Z., Andersen, G. L., & Knight, R. (2010). PyNAST: A flexible tool for aligning sequences to a template alignment. *Bioinformatics*, 26(2), 266–267. <https://doi.org/10.1093/bioinformatics/btp636>
- Cepolina, F., & Razzoli, R. (2024). Review of robotic surgery platforms and end effectors. *Journal of Robotic Surgery*, 18(1), 74. <https://doi.org/10.1007/s11701-023-01781-x>
- Chai, M., Shao, T., Wu, H., Weng, Y., & Zhou, K. (2016). Auto-hair: Fully automatic hair modeling from a single image. *ACM Transactions on Graphics*, 35(4), 1–12. <https://doi.org/10.1145/2897824.2925961>
- Chen, Z., Huang, Z., Gan, Y., Hu, F., Rönning, J., & Li, S. (2025). Development of a 6-DoF robotic manipulator with smart actuators and real-time inverse kinematics. In *2025 International Conference on Networking, Sensing and Control (ICNSC)* (pp. 1–6).
- Choset, H., & Pignon, P. (1998). Coverage path planning: The boustrophedon cellular decomposition. In *Field and Service Robotics* (pp. 203–209).
- Cui, C., Ma, Y., Cao, X., Ye, W., Zhou, Y., Liang, K., Chen, J., Lu, J., Yang, Z., Liao, K.-D., & Gao, T. (2024). A survey on multimodal large language models for autonomous driving. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops* (pp. 958–979).
- Da Costa, D. (2007). *Textured tresses: The ultimate guide to maintaining and styling natural hair*. Simon and Schuster.
- Deng, Z., Luo, H., Gao, X., & Liu, P. (2025). Fusion-Based localization system integrating UWB, IMU, and vision. *Applied Sciences*, 15(12), 6501. <https://doi.org/10.3390/app15126501>
- Dennler, N., Shin, E., Mataric, M., & Nikolaidis, S. (2021). Design and evaluation of a hair combing system using a general-purpose robotic arm. In *Proceedings of the IEEE/RSJ*

- International Conference on Intelligent Robots and Systems (IROS)* (pp. 3739–3746).
- Fraundorfer, F., & Scaramuzza, D. (2012). Visual odometry: Part ii: Matching, robustness, optimization, and applications. *IEEE Robotics & Automation Magazine*, 19(2), 78–90. <https://doi.org/10.1109/MRA.2012.2182810>
- Fu, C., Huang, H., & Weibel, R. (2021). Adaptive simplification of GPS trajectories with geographic context—a quadtree-based approach. *International Journal of Geographical Information Science*, 35(4), 661–688. <https://doi.org/10.1080/13658816.2020.1778003>
- Fu, D., Li, X., Wen, L., Dou, M., Cai, P., Shi, B., & Qiao, Y. (2024). Drive like a human: Rethinking autonomous driving with large language models. In 2024 *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)* (pp. 910–919).
- Gabriely, Y., & Rimón, E. (2001). Spanning-tree based coverage of continuous areas by a mobile robot. *Annals of Mathematics and Artificial Intelligence*, 31(1-4), 77–98. <https://doi.org/10.1023/A:1016610507833>
- Grigorescu, S., Trasnea, B., Cocias, T., & Macesanu, G. (2020). A survey of deep learning techniques for autonomous driving. *Journal of Field Robotics*, 37(3), 362–386. <https://doi.org/10.1002/rob.21918>
- Gronier, J. (1966). Automatic hair-cutting machine having programmed control means for cutting hair in a predetermined style (Patent No. US3241562A).
- Guizilini, V., Lee, K.-H., Ambrus, R., & Gaidon, A. (2022). Learning optical flow, depth, and scene flow without real-world labels. *IEEE Robotics and Automation Letters*, 7(2), 3491–3498. <https://doi.org/10.1109/LRA.2022.3145057>
- Han, B., Qu, T., Tong, X., Jiang, J., Zlatanova, S., Wang, H., & Cheng, C. (2022). Grid-optimized UAV indoor path planning algorithms in a complex environment. *International Journal of Applied Earth Observation and Geoinformation*, 111, 102857. <https://doi.org/10.1016/j.jag.2022.102857>
- Han, D., Mulyana, B., Stankovic, V., & Cheng, S. (2023). A survey on deep reinforcement learning algorithms for robotic manipulation. *Sensors*, 23(7), 3762. <https://doi.org/10.3390/s23073762>
- Harrison, S., & Sinclair, R. (2003). Hair colouring, permanent styling and hair structure. *Journal of Cosmetic Dermatology*, 2(3-4), 180–185. <https://doi.org/10.1111/j.1473-2130.2004.00064.x>
- Horaud, R., & Dornaika, F. (1995). Hand-eye calibration. *The International Journal of Robotics Research*, 14(3), 195–210. <https://doi.org/10.1177/027836499501400301>
- Hua, C., Xu, J., Huang, Z., Liao, B., & Li, S. (2026). Optimization-based finite-time multi-robot formation: A zeroing neurodynamics method. *Tsinghua Science and Technology*, 31(1), 162–179. <https://doi.org/10.26599/TST.2024.9010180>
- Hua, J., Zeng, L., Li, G., & Ju, Z. (2021). Learning for a robot: Deep reinforcement learning, imitation learning, transfer learning. *Sensors*, 21(4), 1278. <https://doi.org/10.3390/s21041278>
- Huang, Z., Li, S., & Rönning, J. (2025). Safety in robotic haircutting. *Authorea*.
- Huang, Z., Zhang, Z., Hua, C., Liao, B., & Li, S. (2024). Leveraging enhanced egret swarm optimization algorithm and artificial intelligence-driven prompt strategies for portfolio selection. *Scientific Reports*, 14(1), 26681. <https://doi.org/10.1038/s41598-024-77925-2>
- Hughes, J., Plumb-Reyes, T., Charles, N., Mahadevan, L., & Rus, D. (2021). Detangling hair using feedback-driven robotic brushing. In 2021 *IEEE 4th International Conference on Soft Robotics (RoboSoft)* (pp. 487–494).
- Ibarz, J., Tan, J., Finn, C., Kalakrishnan, M., Pastor, P., & Levine, S. (2021). How to train your robot with deep reinforcement learning: Lessons we have learned. *The International Journal of Robotics Research*, 40(4-5), 698–721. <https://doi.org/10.1177/0278364920987859>
- International Organization for Standardization. (2014). ISO 13482:2014 robots and robotic devices – safety requirements for personal care robots. Retrieved June 6, 2026, from <https://www.iso.org/standard/53820.html>.
- International Organization for Standardization. (2016). ISO/TS 15066:2016 robots and robotic devices – collaborative robots. Retrieved June 6, 2026, from <https://www.iso.org/standard/62996.html>.
- Jang, E., Irpan, A., Khansari, M., Kappler, D., Ebert, F., Lynch, C., Levine, S., & Finn, C. (2022). BC-Z: Zero-shot task generalization with robotic imitation learning. In *Proceedings of The 5th Conference on Robot Learning* (pp. 991–1002).
- Kavraki, L. E., Kolountzakis, M. N., & Latombe, J.-C. (1998). Analysis of probabilistic roadmaps for path planning. *IEEE Transactions on Robotics and Automation*, 14(1), 166–171. <https://doi.org/10.1109/70.660866>
- Khan, A. T., Cao, X., Li, Z., & Li, S. (2022). Evolutionary computation based real-time robot arm path-planning using beetle antennae search. *EAI Endorsed Transactions on AI and Robotics*, 1(1), 1–10. <https://doi.org/10.4108/airo.v1i.6>
- Khan, A. T., & Li, S. (2026a). CNC-Inspired Robotic hair cutting: A comprehensive survey on precision personal care automation. *Journal of Artificial Intelligence for Automation*, 1(1), 1–14. <https://doi.org/10.53941/jaia.2026.100002>
- Khan, A. T., & Li, S. (2026b). Robotic haircutting systems: A survey of methods, challenges, and hair modeling insights. *IEEE Journal of Selected Areas in Sensors*, 3(2), 93–104. <https://doi.org/10.1109/JSAS.2026.3654480>
- Kim, H., Ohmura, Y., Nagakubo, A., & Kuniyoshi, Y. (2023). Training robots without robots: Deep imitation learning for master-to-robot policy transfer. *IEEE Robotics and Automation Letters*, 8(5), 2906–2913. <https://doi.org/10.1109/LRA.2023.3262423>
- Kim, M. J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., & Vuong, Q. (2024). OpenVLA: An open-source vision-language-action model. *arXiv Preprint arXiv:2406.09246*.
- Kim, S., Kanazawa, N., Hasegawa, S., Kawaharazuka, K., & Okada, K. (2025). Front hair styling robot system using path planning for root-centric strand adjustment. In 2025

- IEEE/SICE International Symposium on System Integration (SII)* (pp. 544–549).
- Kim, U., Lee, D.-H., Yoon, W. J., Hannaford, B., & Choi, H. R. (2015). Force sensor integrated surgical forceps for minimally invasive robotic surgery. *IEEE Transactions on Robotics*, 31(5), 1214–1224. <https://doi.org/10.1109/TRO.2015.2473515>
- Krogus, M., Haggemiller, A., & Olson, E. (2019). Flexible layouts for fiducial tags. In 2019 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 1898–1903).
- Kyosuke, F., Terada, T., & Tsukamoto, M. (2014). A system for supporting self-haircuts using camera equipped robot. In *Proceedings of the 12th International Conference on Advances in Mobile Computing and Multimedia* (pp. 34–42).
- Kyranini, M., Lygerakis, F., Rajavenkatanarayanan, A., Sevastopoulos, C., Nambiappan, H. R., Chaitanya, K. K., Babu, A. R., Mathew, J., & Makedon, F. (2021). A survey of robots in healthcare. *Technologies*, 9(1), 8. <https://doi.org/10.3390/technologies9010008>
- Lamini, C., Benhlima, S., & Elbekri, A. (2018). Genetic algorithm based approach for autonomous mobile robot path planning. *Procedia Computer Science*, 127, 180–189. <https://doi.org/10.1016/j.procs.2018.01.113>
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444. <https://doi.org/10.1038/nature14539>
- Li, Q., Gama, F., Ribeiro, A., & Prorok, A. (2020). Graph neural networks for decentralized multi-robot path planning. In 2020 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 11785–11792).
- Li, S. (2025). *Haircutting robots*. Springer.
- Li, S. (2025). Haircutting robots: From theory to practice. *Automation*, 6(3), 47. <https://doi.org/10.3390/automation6030047>
- Li, Y., Zhu, S., Yu, Y., & Wang, Z. (2018). An improved graph-based visual localization system for indoor mobile robot using newly designed markers. *International Journal of Advanced Robotic Systems*, 15(2), 1729881418769191. <https://doi.org/10.1177/1729881418769191>
- Lin, S., Liu, A., Wang, J., & Kong, X. (2023). An intelligence-based hybrid PSO-SA for mobile robot path planning in warehouse. *Journal of Computational Science*, 67, 101938. <https://doi.org/10.1016/j.jocs.2022.101938>
- Liu, L., Guo, F., Zou, Z., & Duffy, V. G. (2024). Application, development and future opportunities of collaborative robots (cobots) in manufacturing: A literature review. *International Journal of Human-Computer Interaction*, 40(4), 915–932. <https://doi.org/10.1080/10447318.2022.2041907>
- Liu, W., Hu, J., Zhang, H., Wang, M. Y., & Xiong, Z. (2023). A novel graph-based motion planner of multi-mobile robot systems with formation and obstacle constraints. *IEEE Transactions on Robotics*, 40, 714–728. <https://doi.org/10.1109/TRO.2023.3339989>
- Ma, H., Song, A., Li, J., Ge, L., Fu, C., & Zhang, G. (2025). Legged odometry based on fusion of leg kinematics and IMU information in a humanoid robot. *Biomimetic Intelligence and Robotics*, 5(1), 100196. <https://doi.org/10.1016/j.birob.2024.100196>
- Ma, Y., Song, Z., Zhuang, Y., Hao, J., & King, I. (2024). A survey on vision-language-action models for embodied ai. *arXiv Preprint arXiv:2405.14093*.
- Maiolino, P., Maggiali, M., Cannata, G., Metta, G., & Natale, L. (2013). A flexible and robust large scale capacitive tactile system for robots. *IEEE Sensors Journal*, 13(10), 3910–3917. <https://doi.org/10.1109/JSEN.2013.2258149>
- Manish Sinha. (2020). Buzz robotics. Retrieved June 6, 2026, from https://www.youtube.com/watch?v=jY_gpi_gsRI.
- McKinsey & Company. (2020). How COVID-19 is changing the world of beauty. Retrieved June 6, 2026, from <https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/how-covid-19-is-changing-the-world-of-beauty>.
- Moshayedi, A. J., Roy, A. S., Karan, U., Zhang, M., & Bassir, D. (2026). An integrated beetle antennae search-enabled navigation framework for omnidirectional AGV Mobile robots in unknown environments. *EAI Endorsed Transactions on AI and Robotics*, 5(2), 1–20. <https://doi.org/10.4108/airo.11438>
- Muhammad, K., Ullah, A., Lloret, J., Del Ser, J., & De Albuquerque, V. H. C. (2020). Deep learning for safe autonomous driving: Current challenges and future directions. *IEEE Transactions on Intelligent Transportation Systems*, 22(7), 4316–4336. <https://doi.org/10.1109/TITS.2020.3032227>
- Naab, C., & Zheng, Z. (2022). Application of the unscented Kalman filter in position estimation a case study on a robot for precise positioning. *Robotics and Autonomous Systems*, 147, 103904. <https://doi.org/10.1016/j.robot.2021.103904>
- Olson, E. (2011). AprilTag: A robust and flexible visual fiducial system. In 2011 *IEEE International Conference on Robotics and Automation* (pp. 3400–3407).
- Onkarappa, N., & Sappa, A. D. (2014). Speed and texture: An empirical study on optical-flow accuracy in ADAS scenarios. *IEEE Transactions on Intelligent Transportation Systems*, 15(1), 136–147. <https://doi.org/10.1109/TITS.2013.2274760>
- Panasonic. (2011). Panasonic to unveil innovative communication assistance robot “HOSPI-Rimo” and new models of hair-washing robot and “RoboticBed®” at International H.C.R. 2011. Retrieved June 6, 2026, from <https://news.panasonic.com/press/news/official.data/data.dir/en110926-2/en110926-2.pdf>.
- Raja, G., Suresh, S., Anbalagan, S., Ganapathisubramanian, A., & Kumar, N. (2021). PFIN: An efficient particle filter-based indoor navigation framework for UAVs. *IEEE Transactions on Vehicular Technology*, 70(5), 4984–4992. <https://doi.org/10.1109/TVT.2021.3072727>
- Robinson, D. W. (2000). *Design and analysis of series elasticity in closed-loop actuator force control* [Unpublished doctoral dissertation]. Massachusetts Institute of Technology.
- Roumeliotis, S. I., & Bekey, G. A. (2000). Bayesian estimation and Kalman filtering: A unified framework for mobile robot localization. In *Proceedings 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and*

- Automation. Symposia Proceedings (Cat. No. 00CH37065)* (pp. 2985–2992).
- Sklyarova, V., Chelishev, J., Dogaru, A., Medvedev, I., Lempitsky, V., & Zakharov, E. (2023). Neural haircut: Prior-guided strand-based hair reconstruction. In *Proceedings of the IEEE/CVF International Conference Computer Vision (ICCV)* (pp. 19762–19773).
- Stuff Made Here. (2021). I made a hair cutting machine. https://www.youtube.com/watch?v=7zBrbdU_y0s (Accessed on 2026-06-06).
- Sugino, T., Shimoda, Y., Okui, M., & Nakamura, T. (2024). Knee joint assist method focusing on joint viscosity properties using IMU-based exoskeleton assist device. In *2024 16th International Conference on Human System Interaction (HSI)* (pp. 1–7).
- Thimbleby, H. (2003). The directed Chinese postman problem. *Software: Practice and Experience*, 33(11), 1081–1096. <https://doi.org/10.1002/spe.540>
- Tsiakas, K., Tsardoulis, E., & Symeonidis, A. L. (2024). Autonomous full 3D coverage using an aerial vehicle, performing localization, path planning, and navigation towards indoors inventorying for the logistics domain. *Robotics*, 13(6), 83. <https://doi.org/10.3390/robotics13060083>
- Umari, H., & Mukhopadhyay, S. (2017). Autonomous robotic exploration based on multiple rapidly-exploring randomized trees. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 1396–1402).
- Wang, J., & Olson, E. (2016). AprilTag 2: Efficient and robust fiducial detection. In *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 4193–4198).
- Wang, K., Ding, L., Dailami, F., & Matthews, J. (2025). A contemporary review of collaborative robotics employed in manufacturing finishing operations: Recent progress and future directions. *Machines*, 13(9), 772. <https://doi.org/10.3390/machines13090772>
- Williams, G., Drews, P., Goldfain, B., Rehg, J. M., & Theodorou, E. A. (2018). Information-theoretic model predictive control: Theory and applications to autonomous driving. *IEEE Transactions on Robotics*, 34(6), 1603–1622. <https://doi.org/10.1109/TRO.2018.2865891>
- Wolf, S., Grioli, G., Eiberger, O., Friedl, W., Grebenstein, M., Höppner, H., Burdet, E., Caldwell, D. G., Carloni, R., Catalano, M. G., & Lefeber, D. (2016). Variable stiffness actuators: Review on design and components. *IEEE/ASME Transactions on Mechatronics*, 21(5), 2418–2430. <https://doi.org/10.1109/TMECH.2015.2501019>
- Wu, Z., Li, J., Cai, M., Lin, Y., & Zhang, W. (2016). On membership of black-box or white-box of artificial neural network models. In *2016 IEEE 11th Conference on Industrial Electronics and Applications (ICIEA)* (pp. 1400–1404).
- Xu, Z., Zhan, X., Xiu, Y., Suzuki, C., & Shimada, K. (2023). Onboard dynamic-object detection and tracking for autonomous robot navigation with rgb-d camera. *IEEE Robotics and Automation Letters*, 9(1), 651–658. <https://doi.org/10.1109/LRA.2023.3334683>
- Yoo, U., Dennler, N., Nikolaidis, S., Mataric, M., Oh, J., & Ichnowski, J. (2024). Moe-hair: Toward soft and compliant contact-rich hair manipulation and care. In *Companion of the ACM/IEEE International Conference Human-Robot Interaction* (pp. 1163–1167).
- Yoo, U., Dennler, N., Xing, E., & Mataric, M. (2025). Soft and compliant contact-rich hair manipulation and care. In *Proceedings of the ACM/IEEE International Conference Human-Robot Interaction* (pp. 610–619).
- Zakharov, E., Sklyarova, V., Black, M., Nam, G., Thies, J., & Hilliges, O. (2024). Human hair reconstruction with strand-aligned 3D Gaussians. In *Proceedings of the European Conference Computer Vision (ECCV)* (pp. 409–425).
- Zare, M., Kebria, P. M., Khosravi, A., & Nahavandi, S. (2024). A survey of imitation learning: Algorithms, recent developments, and challenges. *IEEE Transactions on Cybernetics*, 54(12), 7173–7186. <https://doi.org/10.1109/TCYB.2024.3395626>
- Zhang, T., Zhang, W., & Gupta, M. M. (2017). Resilient robots: Concept, review, and future directions. *Robotics*, 6(4), 22. <https://doi.org/10.3390/robotics6040022>
- Zhang, W. J., Yang, G., Lin, Y., Ji, C., & Gupta, M. M. (2018). On definition of deep learning. In *2018 World Automation Congress (WAC)* (pp. 1–5). <https://doi.org/10.23919/WAC.2018.8430387>.
- Zhang, Y., Hsiao, M., Dong, J., Engel, J., & Dellaert, F. (2021). MR-iSAM2: Incremental smoothing and mapping with multi-root bayes tree for multi-robot SLAM. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 8671–8678).
- Zhao, G., & Zhu, M. (2020). Pareto optimal multirobot motion planning. *IEEE Transactions on Automatic Control*, 66(9), 3984–3999. <https://doi.org/10.1109/TAC.2020.3025870>
- Zhao, L., Wang, W., He, Q., Yan, L., & Li, X. (2025). Visual-inertial autonomous UAV navigation in complex illumination and highly cluttered under-canopy environments. *Drones*, 9(1), 27. <https://doi.org/10.3390/drones9010027>
- Zhou, Y., Hu, L., Xing, J., Chen, W., Kung, H.-W., Tong, X., & Li, H. (2018). HairNet: Single-view hair reconstruction using convolutional neural networks. In *Proceedings of the European Conference Computer Vision (ECCV)* (pp. 235–251).
- Zhu, K., & Zhang, T. (2021). Deep reinforcement learning based mobile robot navigation: A review. *Tsinghua Science and Technology*, 26(5), 674–691. <https://doi.org/10.26599/TST.2021.9010012>
- Zitkovich, B., Yu, T., Xu, S., Xu, P., Xiao, T., Xia, F., Wu, J., Wohlfahrt, P., Welker, S., Wahid, A., & Vuong, Q. (2023). RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of The 7th Conference on Robot Learning* (pp. 2165–2183).